ModelIt!

Teacher's Guide

# Powered by the Sun

How Pizza, Basketball, and Your Heartbeat Are All Solar Energy

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 5 |
| **NGSS Standard** | 5-PS3-1 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: 5-PS3-1**

Use models to describe that energy in animals' food was once energy from the sun.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop a model to trace energy flow from sun to organisms. |
| **Disciplinary Core Idea** | **PS3.D Energy in Chemical Processes** — Energy from the sun is transferred to organisms through photosynthesis. |
| **Crosscutting Concept** | **Energy and Matter** — Students trace how energy flows through systems. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A plant in a dark room will eventually die because it cannot perform photosynthesis. What happens to animals that depend on this plant?

**CONSTRUCTED RESPONSE:**

Explain how the energy in a hamburger can be traced back to the sun.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

All life on Earth runs on solar energy captured through photosynthesis. This fundamental process converts light energy into chemical energy.

## Photosynthesis Basics

Plants use chlorophyll to capture light energy. This energy powers the reaction that combines CO2 and water to produce glucose (sugar) and oxygen.

## The 10% Rule

Only about 10% of energy is transferred between trophic levels. The rest is lost as heat. This explains why food chains rarely have more than 4-5 levels.

## Temperature Effects

Photosynthesis is a chemical reaction that proceeds faster at moderate temperatures. Too cold or too hot reduces efficiency.

## Seasonal Changes

Winter reduces both sunlight intensity and day length, dramatically decreasing photosynthesis and available food energy.

# 4 The LEVER Framework

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**About LEVER**

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L — Locate the System**
Students identify sunlight, temperature, photosynthesis rate, and energy available to animals as key components.

**E — Establish Relationships**
Students establish that sunlight and temperature positively affect photosynthesis, which positively affects available energy.

**V — Visualize & Model**
Students build the energy flow model showing the solar energy chain.

**E — Evaluate Outcomes**
Students run winter and eclipse scenarios to see the cascade effect on animal energy.

**R — Revise & Extend**
Students add water or CO2 to make the model more complete.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Everything you do is powered by the sun."*

**Watch For**

Ask students what they ate for breakfast, trace it back.

## Slide 2: Learning Objectives

2 min

**Say:** *"Today we trace energy from sun to you."*

**Watch For**

Have students predict the chain.

## Slide 3: Big Question

2 min

**Say:** *"How does sunlight become your energy?"*

**Watch For**

Quick partner discussion.

## Slide 4: LEVER Framework

2 min

**Say:** *"We'll use LEVER to build our model."*

**Watch For**

Review each step briefly.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Let's identify the parts of our energy system."*

**Watch For**

Guide sorting of external vs internal.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"All three relationships are positive. Why?"*

**Watch For**

Discuss why energy flows one direction.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"What happens in winter? Let's test it."*

**Watch For**

Have students run winter scenario.

## Slide 8: Discoveries

5 min

**Say:** *"This explains why animals hibernate!"*

**Watch For**

Connect to real animal behaviors.

## Slide 9: STEM Challenge

10 min

**Say:** *"Design a greenhouse for Mars astronauts."*

**Watch For**

Distribute design materials.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Sunlight Intensity, Temperature
- **INTERNAL:** Photosynthesis Rate, Energy Available to Animals

## Relationships

- **Sunlight Intensity to Photosynthesis Rate:** POSITIVE (+) — More sunlight provides more energy for photosynthesis.
- **Temperature to Photosynthesis Rate:** POSITIVE (+) — Warmer temperatures speed up the chemical reactions.
- **Photosynthesis Rate to Energy Available:** POSITIVE (+) — More photosynthesis produces more plant sugar for animals to eat.

## Key Discoveries

- All energy in food originally came from sunlight
- Photosynthesis converts solar energy into chemical energy (sugar)
- When sunlight drops, the entire food chain is affected
- Animals hibernate and migrate because winter reduces available energy

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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**Why This Matters**

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
| --- | --- |
| "Plants get energy from soil" | Plants get MINERALS from soil but ENERGY from sunlight through photosynthesis. |
| "Animals make their own energy" | Animals get ALL their energy from food, which traces back to photosynthesis. |
| "The sun is optional for life" | With few exceptions (deep sea vents), all life depends on solar energy. |